

Rolling the Dice on Destiny: Predicting the next Best Board Game

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Link to GitHub for the code: [Lenarbrs/2026 Board Game Prediction](https://github.com/Lenarbrs/2026_Board_Game_Prediction)

1. Introduction

The board game industry has experienced remarkable growth over the past two decades, with thousands of new games released each year and an increasingly diverse player base. Platforms such as BoardGameGeek (BGG) have become central repositories for game ratings, reviews, and rankings, providing a rich source of data for understanding the factors that contribute to a game's success.

Bayesian statistics provide a way to analyse this study subject by allowing prior beliefs about game design factors to be combined with observed data, producing posterior distributions that directly quantify the probability of different outcomes. This framework is particularly well suited for board game analysis, where relationships between game mechanics, complexity, and player appreciation are inherently uncertain and may evolve over time. In this project, we aim to investigate which game characteristics are most strongly associated with highly rated board games and to estimate the probability that a future game could achieve outstanding success. More specifically, we examine whether game complexity and the presence of popular mechanics significantly increase the likelihood of a game being the best rated game of the year.

Using a Bayesian regression model, we will estimate the influence of these design features while accounting for uncertainty in the observed data. The resulting posterior distributions will not only help explain what makes a game successful but will also provide a probabilistic framework for predicting the characteristics of the best rated game of the year, for the next year. Ultimately, this analysis seeks to answer a practical question for designers and publishers alike: *what combination of features maximizes the probability of creating a highly acclaimed board game?* Or, in Bayesian words: *Is it possible to predict next year's best rated board games knowing the rating of board games published in the past?*

Hypothesis 1: Impact of Complexity and Mechanics

Does the combination of game complexity and specific mechanics significantly increase the probability of a board game receiving a high average rating?

The popular mechanics are selected as the top 10 most frequent mechanics on BGG's database.

- **H0:** The posterior coefficients associated with game complexity and selected mechanics are centered around zero. These variables have no credible impact on a game's average rating.
- **H1:** The posterior coefficients associated with game complexity and selected mechanics are positive. More complex games and certain mechanics are associated with higher average ratings.

Hypothesis 2: Temporal Evolution of Board Game Quality

Has the average quality of board games increased over time?

- **H0:** The coefficient associated with publication year (β_{year}) is centered around zero. Publication year has no credible effect on a game's average rating.
- **H1:** The coefficient associated with publication year (β_{year}) is positive. More recently published games tend to receive higher ratings than older games.

This second hypothesis is particularly relevant for forecasting a potential 2026 Board Game of the Year, as it tests whether there is a long-term upward trend in player evaluations that should be incorporated into future predictions.

2. Problem History

The modern board game industry has experienced substantial growth since the mid-1990s. The release of *The Settlers of Catan* (Klaus Teuber, 1995) can be identified as a major turning point in the development of modern hobby board games¹. Unlike traditional mass-market games, *Catan* emphasized strategic decision-making, resource management, and player interaction, helping establish what is now commonly referred to as the “modern board game renaissance.”

During the 2000s and 2010s, the industry expanded rapidly through online communities, specialized publishers, and crowdfunding platforms such as Kickstarter². Crowdfunded games were found to be more innovative on average than traditionally published games and often influenced future game development trends. At the same time, BGG³, founded in 2000, became the dominant online platform for board game ratings, rankings, reviews, and metadata with more than 150,000 listed games and millions of user ratings. An important characteristic of BGG is that its ranking system does not rely solely on raw average ratings. To avoid situations in which a game with only a few extremely positive ratings would artificially appear among the highest-ranked games, BGG incorporates a modified score called the “Geek Rating.” According to BGG documentation⁴, the ranking algorithm introduces a number of artificial or “dummy” ratings before calculating the final ranking score. This procedure shrinks ratings toward a global average when the number of observations is small.

From a statistical perspective, this approach resembles Bayesian regularization or a Bayesian average. Prior information, represented by the dummy votes, is combined with observed user ratings to obtain a more stable estimate of a game's underlying quality. This principle is closely related to Bayesian inference, where prior beliefs are updated using observed evidence. Analyses of the BGG ranking methodology have shown that the system effectively behaves like a Bayesian averaging process designed to reduce uncertainty associated with small sample sizes.

This historical evolution of both the board game industry and ranking methodologies directly motivates the use of Bayesian statistical tools in our project. Since our objective is to predict the characteristics of a potential Board Game of the Year in 2026, it requires a framework capable of incorporating prior information, handling uncertainty, and producing probabilistic forecasts. The Bayesian paradigm naturally aligns with the logic already embedded in modern board game ranking systems such as BoardGameGeek.

3. Methodology

a. Data Preprocessing

We use the Board Games Dataset published by Melissa Monfared and available on Kaggle⁵, which contains information on all ranked board games on BGG as of February 2021, or 20,327 distinct board games. The

¹ [The History of Catan: From 1995 to 2026 — Catan Almanac](#)

² [Kickstarter](#)

³ [BoardGameGeek | Gaming Unplugged Since 2000](#)

⁴ [ratings | Wiki | BoardGameGeek](#)

⁵ [Board Games](#)

dataset includes several game characteristics, such as the year of publication, minimum and maximum number of players, minimum recommended age, average user rating, number of user ratings, BGG rank, average complexity, number of users who own the game, as well as game domains and mechanics. The BGG rank is a ranking system computed by BoardGameGeek that reflects the overall popularity and appreciation of a game within the board gaming community. Games with lower ranks are generally considered more successful and highly regarded by players.

To prepare the data for our Bayesian analysis, we first created dummy variables for all eight game domains: Wargames, Strategy Games, Family Games, Thematic Games, Abstract Games, Children's Games, Party Games, and Customizable Games. This transformation was necessary because a single game may belong to multiple domains simultaneously. We applied the same approach to game mechanics. However, the dataset contains 182 distinct mechanics, making it impractical to include all of them in our model. To reduce dimensionality while retaining the most informative features, we selected the 10 most frequent mechanics in the dataset: Dice Rolling, Hand Management, Set Collection, Variable Player Powers, Hexagon Grid, Simulation, Card Drafting, Tile Placement, Modular Board, and Area Majority / Influence (see *Figure 1*).

Several data-cleaning steps were then performed to improve data quality. During this process, we decided to be strict because the sample was sufficiently large to allow us to remove observations with uncertain values. We removed games with a reported play time equal to zero or greater than 540 minutes (9 hours), as such values are unusually high and may correspond to data-entry errors or highly atypical games. Similarly, observations with uncertain or implausible values such as a maximum number of players equal to 99 or a minimum or maximum number of players equal to 0 were discarded. Because our dataset stops at 2021, our objective is to predict the characteristics of a potential 2022 best rated board game. Hence, we also restricted the analysis to relatively modern games. Games published before 1850 were removed, as they are unlikely to be representative of contemporary board game design, as well as because there is uncertainty in values before this threshold. Eventually, we removed all observations featuring NAs.

After applying these preprocessing and filtering procedures, the final dataset contains 9,139 unique board games. The distributions of the continuous and categorical variables are presented in Figures 1 and 2, respectively.

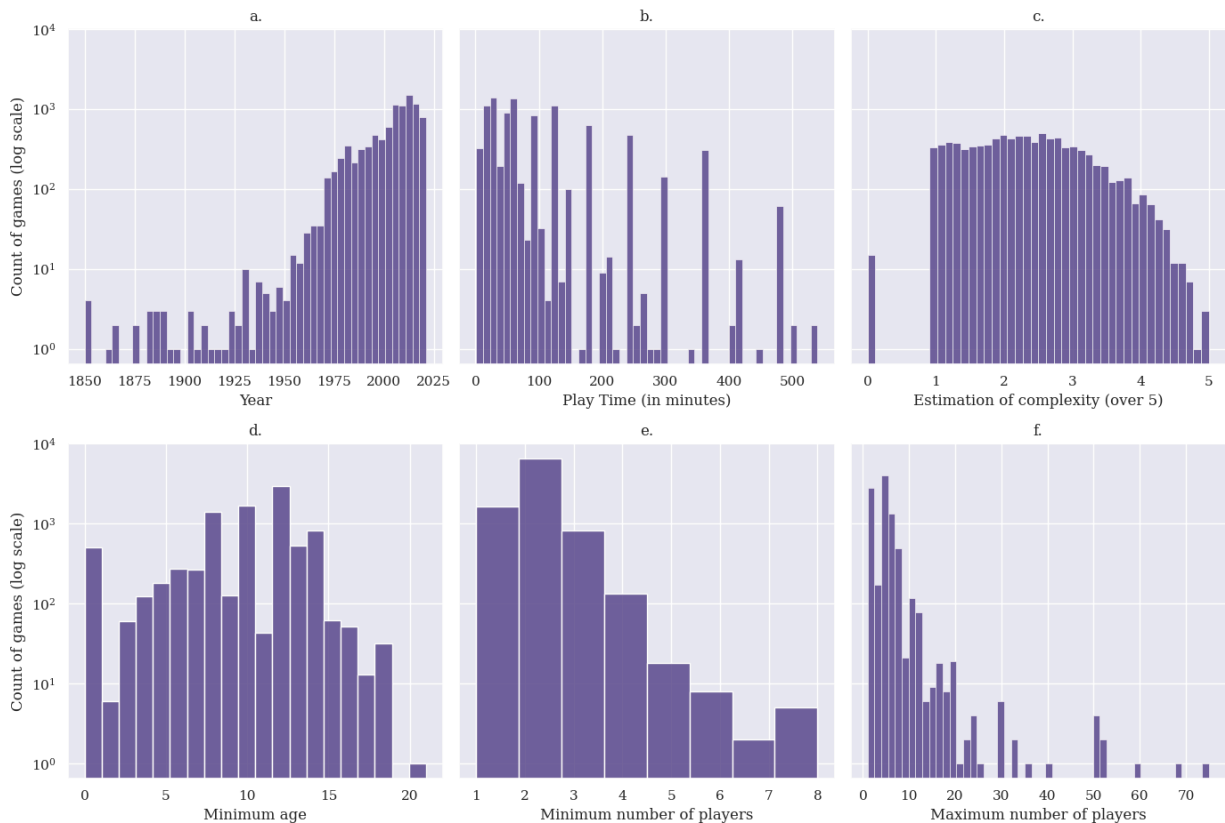


Figure 1. Distribution of continuous variables

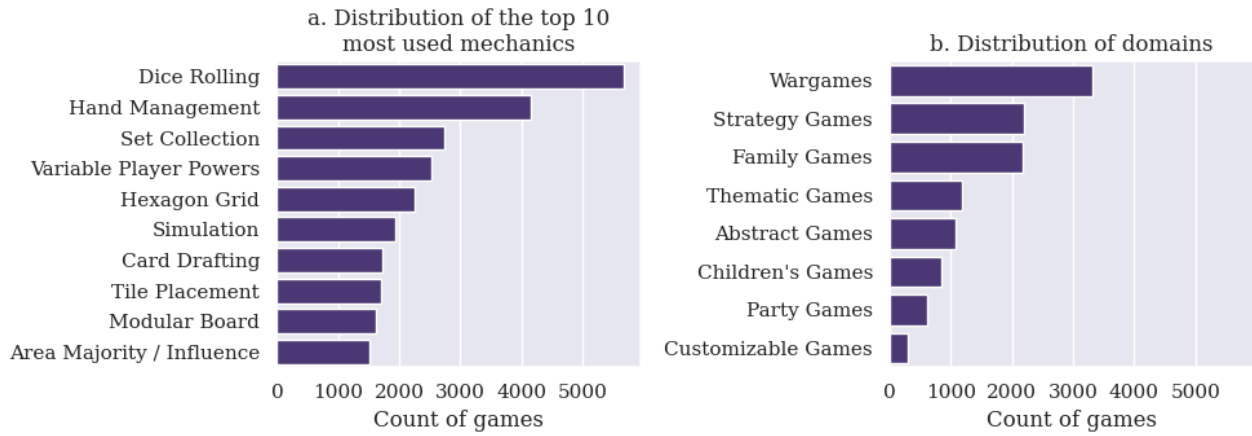


Figure 2. Distribution of categorical variables

Our objective is to identify the characteristics of the best rated board game of 2021 or 2022. Before building a Bayesian predictive model, we first explored the distribution of game ratings and their evolution over time (see *Figure 3*). The distribution is concentrated around ratings between 6 and 8, with only a relatively small proportion of games achieving exceptionally high scores. This suggests that obtaining a rating above 8.0 is uncommon and may therefore serve as a meaningful indicator of outstanding game quality. We then examined the average ratings of award-winning games over time. Award-winning games tend to receive increasingly higher ratings in more recent years. This trend could indicate that the standards required to achieve critical and commercial success have risen over time, with modern award winners generally being more highly appreciated by players than earlier winners.

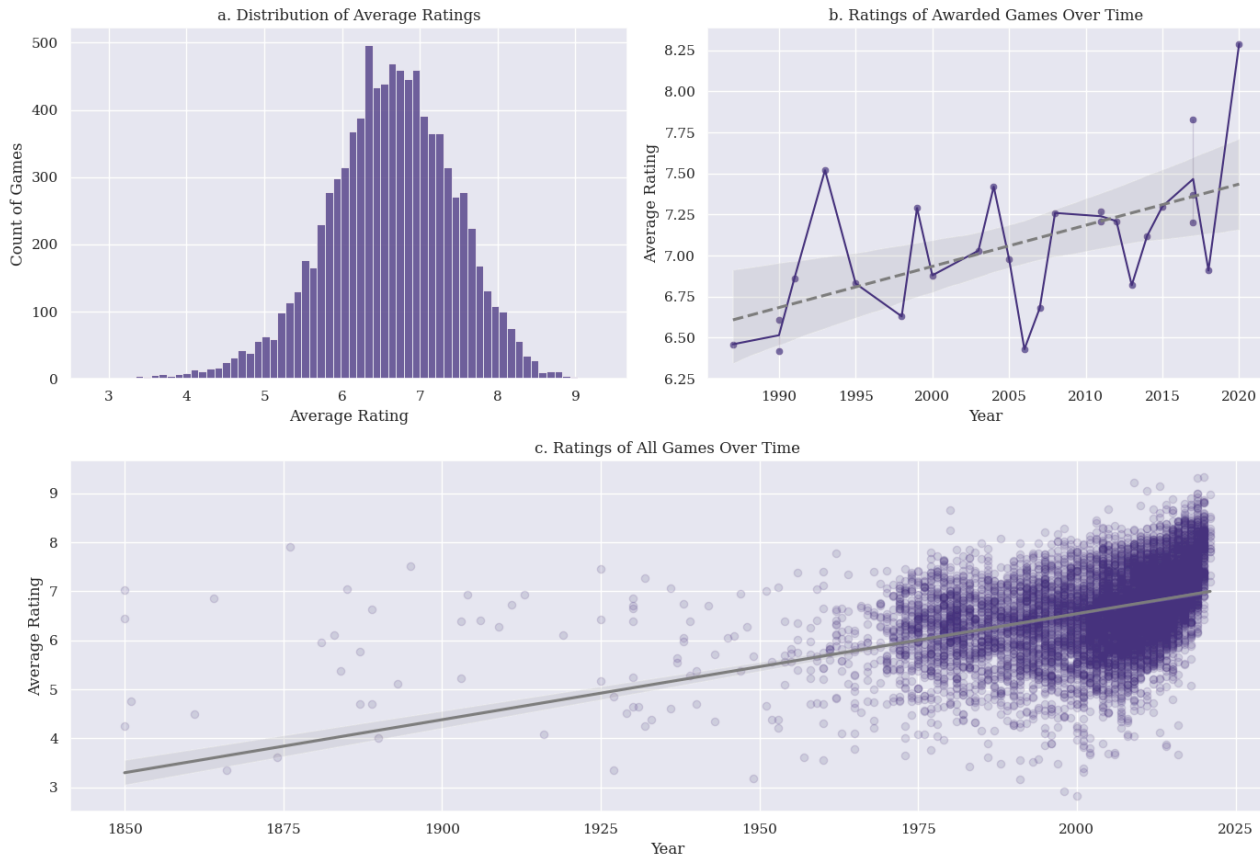


Figure 3. Distribution of ratings

However, this increase is not unique to award-winning games, the average rating of all games in the dataset has also increased over time. Several factors may explain this phenomenon. First, the board game industry has become increasingly competitive, encouraging publishers and designers to produce higher-quality games. Second, modern games often benefit from decades of accumulated design knowledge, allowing designers to refine and improve successful mechanics. Finally, the growth of online communities such as BoardGameGeek has made player feedback more accessible, enabling developers to better understand player preferences and expectations. More generally, these preliminary findings highlight that a game's rating is likely influenced by a combination of temporal effects, game complexity, domains, and mechanics, motivating the use of a Bayesian model to quantify the contribution associated with each factor.

b. Bayesian Statistical Framework

To predict the characteristics of the best rated board game of 2021 or 2022, we adopted a Bayesian statistical approach. Unlike traditional frequentist methods, Bayesian statistics treats model parameters as random variables and updates prior beliefs using observed data through Bayes' theorem. This framework allows uncertainty to be quantified directly through probability distributions rather than single-point estimates. Bayesian methods are particularly appropriate for our research question because board game success depends on multiple uncertain and interacting factors, such as complexity, mechanics, domains, and publication year. Bayesian inference therefore enables us to estimate not only the direction of an effect but also the probability that this effect is positive or negative.

→ **Bayesian linear regression**⁶

Bayesian linear regression extends classical linear regression by treating model parameters as random variables rather than fixed unknown quantities. Although the theoretical foundations come from Bayes' theorem, the practical development of Bayesian regression accelerated during the second half of the twentieth century as computational resources became available. Unlike ordinary least squares regression, Bayesian regression produces an entire probability distribution for each coefficient rather than a single estimate. This is particularly useful for our project because our goal is not only to estimate the effect of a game mechanic, but also to quantify our uncertainty regarding that effect.

→ **Student-t regression**⁷

Rather than assuming that game ratings follow a normal distribution around the regression line, we modeled the response variable using a Student-t distribution. Board game ratings may contain extreme observations, such as exceptionally successful games or poorly received titles. A normal likelihood is sensitive to such outliers, whereas the heavier tails of the Student-t distribution make the model more robust. This approach reduces the influence of unusual observations while preserving the overall structure of the data.

This distribution was introduced in 1908 by William Sealy Gosset, a statistician working for the Guinness Brewery under the pseudonym "Student". The Student-t distribution was originally developed to analyze small samples where the population variance was unknown. Compared to the normal distribution, it has heavier tails, making it less sensitive to extreme observations. This property is particularly relevant for board game ratings because a small number of games may receive exceptionally high or low ratings. Using a Student-t likelihood prevents these observations from exerting excessive influence on the model.

→ **MCMC**⁸

The posterior distribution of our Bayesian model cannot be calculated analytically because of the large number of parameters included in the regression. To overcome this problem, we use Markov Chain Monte

⁶ [CSC 411 Lecture 19: Bayesian Linear Regression](#), [Bayesian Linear Regression - GeeksforGeeks](#)

⁷ [Theoretical properties of Bayesian Student-t linear regression - ScienceDirect](#)

⁸ [Markov chain Monte Carlo \(MCMC\) - GeeksforGeeks](#)

Carlo (MCMC) methods. The origins of MCMC can be traced back to the Metropolis algorithm, developed in 1953 by Nicholas Metropolis and collaborators while studying particle interactions in statistical physics. The method generates samples from complex probability distributions through a stochastic process known as a Markov chain. The development of MCMC revolutionized Bayesian statistics because it allowed researchers to estimate models that were previously computationally impossible.

→ No-U-Turn Sampler⁹

The Bayesian model was implemented using PyMC¹⁰, an open-source probabilistic programming library for Python. PyMC allows complex Bayesian models to be expressed using intuitive mathematical syntax while automatically handling posterior inference through advanced sampling algorithms. One of the main advantages of PyMC is its integration of state-of-the-art Markov Chain Monte Carlo samplers, including the No-U-Turn Sampler (NUTS), which enables efficient estimation of high-dimensional Bayesian models.

The sampling algorithm used by PyMC is the No-U-Turn Sampler (NUTS), introduced by Matthew Hoffman and Andrew Gelman in 2014. NUTS is an extension of Hamiltonian Monte Carlo designed to remove one of HMC's main limitations: the need to manually specify the trajectory length. Choosing an inappropriate trajectory can lead either to inefficient exploration or unnecessary computational cost. NUTS automatically adapts the trajectory length during sampling and stops when it detects that the trajectory is starting to double back on itself, hence the term "No-U-Turn". Today, NUTS is considered the state-of-the-art sampling algorithm for Bayesian regression models and is the default sampler in modern probabilistic programming frameworks such as PyMC and Stan.

c. Model

To ensure scale comparability and numerical stability during Markov Chain Monte Carlo (MCMC) sampling, continuous variables (play time, complexity, minimum age, minimum and maximum players) were standardized to zero mean and unit variance (z-scores).

$$X_{scaled} = \frac{X - \bar{X}}{s_X}$$

To capture the temporal evolution of the board game industry, publication year was transformed into a time offset centered on our target baseline, $t_i = Year_i - 2025$. Categorical features (the 10 most frequent mechanics and game domains) were encoded as binary indicators.

Standard regression assumes the impact of game characteristics remains constant. To account for the evolution of the features' impact on ratings over time, the model introduces dynamic feature-time interactions, estimating both a baseline effect and an annual trend for every predictor.

The expected rating μ_i for game i is modeled as:

$$\mu_i = \alpha + \beta_{year} t_i + \sum_{k \in C} (\beta_k + \gamma_k t_i) X_{k,i} + \sum_{j \in D} (\beta_j + \gamma_j t_i) X_{j,i} + \sum_{m \in M} (\beta_m + \gamma_m t_i) X_{m,i}$$

Where C , D , and M represent the sets of Continuous variables, Domains, and Mechanics.

This formulation allows the influence of any feature to vary smoothly through time:

- β (Baseline Effect): The impact of the feature strictly in 2021 (when $t_i = 0$).

⁹ [The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte-Carlo](#)

¹⁰ [Home — PyMC project website](#)

- γ (Temporal Velocity): The annual rate of change in the feature's influence. A positive γ indicates an emerging trend, while a negative γ indicates a declining fad.
- β_{year} : The general temporal drift (inflation or deflation) of overall board game ratings.

Because the target variable (Average Rating) is empirically normally distributed, the observation likelihood is modeled with a Gaussian distribution:

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

Bayesian inference requires specifying prior distributions reflecting pre-data beliefs. We assigned independent, weakly informative priors to allow the data to dominate the posterior while regularizing extreme values.

- The intercept was assigned a Normal prior centered on the average rating observed in the dataset (where $\bar{y}$ is the mean board game rating).

$$\alpha \sim \mathcal{N}(\bar{y}, 1)$$

- Time trends:

$$\beta_{year} \sim \mathcal{N}(0, 0.5)$$

- Feature Baselines: Normal priors were assigned to continuous, domain, and mechanic base coefficients.

$$\beta_k, \beta_j, \beta_m \sim \mathcal{N}(0, 1.0)$$

- Temporal Velocities: tighter priors were assigned to trend shifts, as annual changes in player preferences are assumed to be gradual.

$$\gamma_k, \gamma_j, \gamma_m \sim \mathcal{N}(0, 0.2)$$

- Residual Error: bounded at zero using a Half-Normal distribution.

$$\sigma \sim \text{HalfNormal}(1)$$

The posterior distribution was estimated using MCMC sampling via the No-U-Turn Sampler (NUTS) implemented in PyMC. Four independent chains were run in parallel.

Model convergence was assessed using the Gelman-Rubin statistic ($\hat{R}$), with values strictly below 1.01 indicating that all chains converged to the same posterior distribution.

Statistical credibility for each parameter was evaluated using the 94% Highest Density Interval (HDI).

4. Results

The posterior estimates of the dynamic Bayesian regression model are summarized in *Table 1*. By isolating the baseline effect in 2021 from the annual trend, the model reveals not only what current players value but how those preferences are actively shifting over time. Effects are considered statistically credible if their 94% Highest Density Interval (HDI) excludes zero. We found $\hat{R}$ of 1.0 for all variables.

The baseline coefficients represent the immediate impact of a feature on a game's expected rating for the year 2021. Customizable Games emerged as the strongest domain, granting an estimated 0.420-point increase to a game's baseline rating. This is closely followed by Complexity Average. Because continuous variables were standardized, the 0.363 coefficient indicates that for every one standard deviation increase in a game's complexity, the expected rating increases by over a third of a point. Party Games (0.267) and core mechanics

like Card Drafting (0.182), Dice Rolling (0.148), and Variable Player Powers (0.136) also demonstrated highly credible positive impacts.

Conversely, certain features severely penalize a game's rating. The Roll / Spin and Move mechanic is the most heavily penalized feature overall (-0.307), reflecting the modern hobbyist community's rejection of low-agency, high-randomness movement. Children's Games (-0.280) and Abstract Games (-0.156) also suffer from significant baseline penalties. Additionally, higher Minimum Player counts negatively impact ratings (-0.081), likely reflecting the premium placed on accessibility and lower player-count flexibility in recent years. Notably, general game attributes such as Play Time, Max Players, and Min Age failed to reach statistical significance. Broad domains like Strategy Games and Wargames also showed no credible baseline impact, suggesting that a game's success in these categories is entirely dependent on its specific mechanics and complexity rather than the domain label itself.

The temporal velocity coefficients expose the changing meta of the board game industry, highlighting which features are gaining or losing momentum. Several features are not only highly rated in 2021 but are becoming increasingly valuable every year. Customizable Games exhibits the steepest growth trajectory (+0.042 points per year), indicating a surging demand for modular, expandable content. Mechanics like Card Drafting (+0.008 per year) and Dice Rolling (+0.007 per year) also show significant, continuous upward momentum.

Complexity Average provides a massive baseline boost in 2021, but its annual trend is statistically insignificant. This demonstrates that while heavy games are consistently rated higher, the premium awarded for complexity has plateaued and is no longer inflating. Mechanics like Variable Player Powers and Simulation follow this same pattern: they may be valuable, but their influence remains static over time.

Finally, the model identifies declining fads. Several domains are experiencing a slow but credible decline in favorability. Thematic Games (-0.006 per year) and Abstract Games (-0.004 per year) are actively losing rating potential over time. Most notably, while Family Games do not carry a significant baseline penalty in 2021, their negative trajectory (-0.007 per year) suggests that the community is becoming increasingly critical of this category as the years progress.

figure 4 and figure 5 are graphical descriptions of the results and can be found in the appendix.

5. Conclusion

The initial objective of this research was to predict the characteristics of the best rated board game of the coming years using historical data. The Bayesian dynamic linear regression model effectively addressed this problem by isolating the baseline impact of game features from their temporal trajectories. Instead of treating game mechanics and domains as static elements whose value remains fixed, the model quantifies how player preferences evolve smoothly over time. This probabilistic framework allows for a nuanced understanding of game design, explaining not only what is currently successful, but what is actively gaining momentum within the player community.

The data reveals that the optimal combination for a highly acclaimed board game heavily favors high complexity and customizable formats. To maximize the probability of achieving an outstanding average rating, a game should incorporate mechanics that are currently trending upward, such as Card Drafting and Dice Rolling. Conversely, the data strongly indicates that designers should avoid high-randomness, low-agency mechanics like Roll/Spin and Move, as well as domains like Children's or Abstract games, all of which carry significant baseline penalties. Therefore, predicting or designing the next top-rated game requires synthesizing these highly weighted positive attributes and upward trends while actively avoiding statistically penalized features.

From an external perspective, the model successfully captures the dynamic nature of user ratings but remains constrained by its dataset and structural assumptions. The reliance on BoardGameGeek's specific user base introduces an inherent selection bias, as the platform's demographic skews heavily toward hobbyist gamers who naturally reward complexity and penalize Family or Children's games. Additionally, the model assumes linear temporal trends for game features, which may oversimplify the potentially cyclical or abrupt nature of gaming fads. Finally, while the Bayesian framework robustly quantifies uncertainty and identifies strong correlations, these relationships are not strictly causal; combining highly rated mechanics into a single game will not automatically guarantee a successful design if the underlying execution is flawed.

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7. Appendix

Table 1: Posterior estimates of the Dynamic Bayesian Linear Regression predicting board game ratings.

Feature	Category	Coeff 2021	Significativity Coeff	Yearly Trend	Significativity Trend
Customizable Games	Domain	0.420	Yes	0.042	Yes

Complexity Average	Continuous (Std)	0.363	Yes	0.001	No
Party Games	Domain	0.267	Yes	0.011	Yes
Card Drafting	Mechanic	0.182	Yes	0.008	Yes
Dice Rolling	Mechanic	0.148	Yes	0.007	Yes
Variable Player Powers	Mechanic	0.136	Yes	0.004	No
Hexagon Grid	Mechanic	0.126	Yes	0.005	Yes
Simulation	Mechanic	0.085	Yes	-0.003	No
Wargames	Domain	0.046	No	0.003	No
Strategy Games	Domain	0.030	No	0.001	No
Tile Placement	Mechanic	0.027	No	0.004	No
Play Time	Continuous (Std)	0.023	No	0.001	No
Hand Management	Mechanic	0.022	No	-0.001	No
Modular Board	Mechanic	-0.005	No	-0.000	No
Max Players	Continuous (Std)	-0.008	No	-0.000	No
Min Age	Continuous (Std)	-0.011	No	-0.001	No
Set Collection	Mechanic	-0.029	No	0.000	No
Family Games	Domain	-0.050	No	-0.007	Yes
Min Players	Continuous (Std)	-0.081	Yes	-0.003	Yes
Thematic Games	Domain	-0.140	Yes	-0.006	Yes

Abstract Games	Domain	-0.156	Yes	-0.004	Yes
Children's Games	Domain	-0.280	Yes	0.002	No
Roll / Spin and Move	Mechanic	-0.307	Yes	0.001	No

Figure 4 - Features' base impact & temporal velocity (in 2021)

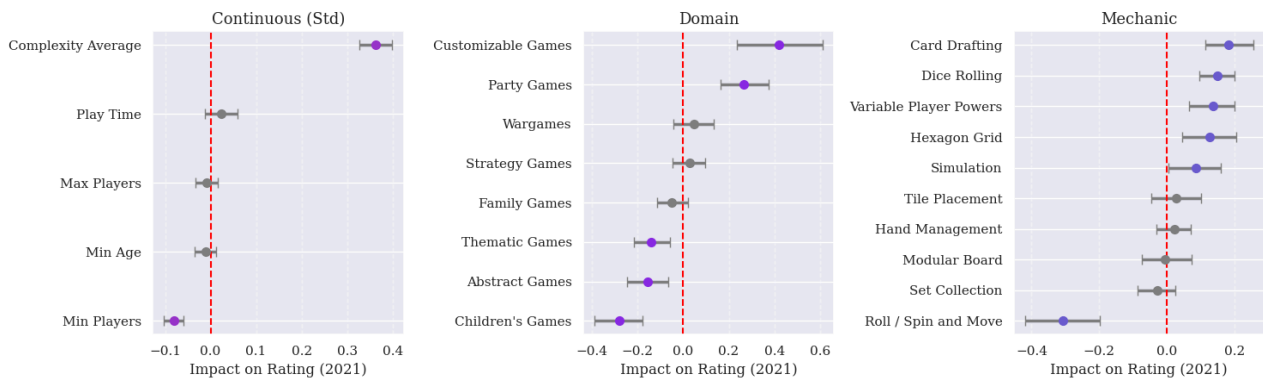


Figure 5 : Statistical Certainty of 2021 Impacts (94% HDI)